Research Project on Linear Algebra

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August 10th, 2021

Statement

The main results of our research project are:

Theorem A

The conjugate classes of the solution of

$$A^k = I_n$$

in $M_n(F)$ *is ....*

Theorem B

The ring homomorphisms of

$$U_n(F) \to F[t]$$

are

1. $g(A) = \cdots$, *where* $A = (a_{ij})$;
2. *for* $i = 1, \ldots, n$, $f_i(A) = \cdots$, *where* $A = (a_{ij})$.

Here we write $A = (a_{ij}) \in U_n(F)$.

Proof of Theorem A

Let F be a field with $\operatorname{char} F = 0$. Let A be a matrix in $M_{m,n}(F)$.

Denote by $B = A^2$. Denote by S the set

$$\{1, 2, 3, \dots\}$$

Let f be a F-linear map

$$\begin{aligned} f\colon V &\longrightarrow W \\ v &\longmapsto w. \end{aligned}$$

Denote by

$$A=\begin{pmatrix} 1 & 2 & \cdots & n \\ 2 & 3 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ n & 1 & \cdots & n-1 \end{pmatrix}, \quad B=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{m,n} \end{bmatrix}.$$

$$C=\begin{bmatrix} 1 & 2 & \cdots & n \\ & 1 & \cdots & n-1 \\ & & \ddots & \vdots \\ & & & 1 \end{bmatrix}, \quad D=\begin{vmatrix} 1 & 2 & \cdots & n \\ 2 & 3 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ n & 1 & \cdots & n-1 \end{vmatrix} =?$$

Denote by

$$A=\begin{pmatrix} 1 & 2 & \cdots & n \\ 2 & 3 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ n & 1 & \cdots & n-1 \end{pmatrix}, \quad B=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{m,n} \end{bmatrix}.$$

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Vector is a kind of special matrix.

α	\alpha	θ	\theta	o	o	υ	\upsilon
β	\beta	ϑ	\vartheta	π	\pi	ϕ	\phi
γ	\gamma	ι	\iota	ϖ	\varpi	φ	\varphi
δ	\delta	κ	\kappa	ρ	\rho	χ	\chi
ϵ	\epsilon	λ	\lambda	ϱ	\varrho	ψ	\psi
ε	\varepsilon	μ	\mu	σ	\sigma	ω	\omega
ζ	\zeta	ν	\nu	ς	\varsigma		
η	\eta	ξ	\xi	τ	\tau		
Γ	\Gamma	Λ	\Lambda	Σ	\Sigma	Ψ	\Psi
Δ	\Delta	Ξ	\Xi	Υ	\Upsilon	Ω	\Omega
Θ	\Theta	Π	\Pi	Φ	\Phi		
$\varGamma$	\varGamma	$\varLambda$	\varLambda	$\varSigma$	\varSigma	$\varPsi$	\varPsi
$\varDelta$	\varDelta	$\varXi$	\varXi	$\varUpsilon$	\varUpsilon	$\varOmega$	\varOmega
$\varTheta$	\varTheta	$\varPi$	\varPi	$\varPhi$	\varPhi		

See lshort.pdf Chapter 4 for more symbols.



Figure: An apple

$$\det, \operatorname{sgn}, \overline{fg}, \oplus, \times, \pm, \mp, \leq, \geq, \xrightarrow{\sim}, \cong, \in, \ni, \subset, \supset, \subseteq, \subsetneq$$
$$, \cup, \cap, \int, \iint, \iiint, \Longrightarrow, \Longleftrightarrow, \hbar, \nabla.$$

$$\det, \operatorname{sgn}, \overline{fg}, \oplus, \times, \pm, \mp, \leq, \geq, \xrightarrow{\sim}, \cong, \in, \ni, \subset, \supset, \subseteq, \subsetneq$$
$$, \cup, \cap, \int, \iint, \iiint, \Longrightarrow, \Longleftrightarrow, \hbar, \nabla.$$

$$\lim_{n\to\infty} a_n = a, \quad \varinjlim_{n\to\infty} X_n, \quad \varprojlim_{n\to\infty} X_n, \quad \dot{a}, \ \ddot{b}.$$

Here is an equation we may cite:

$$\frac{a^{b^c}}{c_{b_a}}, \quad \sum_{i=1}^{n} a_i, \quad \prod_{i,j=1}^{+\infty} X_i^j, \quad \text{some text here.} \tag{2.1}$$

Lemma

I have an apple, I have a pen.

Proof.

See Eq. (2.1), I got an applepen. □